



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

– Hira Kuma

IP Addressing



IP Address = Combination of two IDs: Network ID and Host ID

i.e. IP address $\rightarrow$ Network id + Host id

❖ What is network id ?

- Network id is the part of an IP address that identifies a particular network. It tells a device which network the destination IP belongs to.
- Example : In **Class C** ip address
IP address : 192.168.1.10
Subnet mask : 255.255.255.0
Then
Network id : 192.168.1.0 (here, 192.168.1.0 identifies the network, while 10 identifies the host within that network)

❖ What is host id ?

- Host id is the part of an IP address that identifies a particular device (host) within a network.
- Example : In **Class C** ip address
IP address : 192.168.1.10
Subnet mask : 255.255.255.0

Then

Network id : 192.168.1.0

Host id : 10

So, here

Network id → Which network ?

Host id → Which device in that network ? (logically)

Class A 255.0.0.0 or /8
0.0.0.0 - 127.0.0.0



Class B 255.255.0.0 or /16
128.0.0.0 - 191.255.0.0

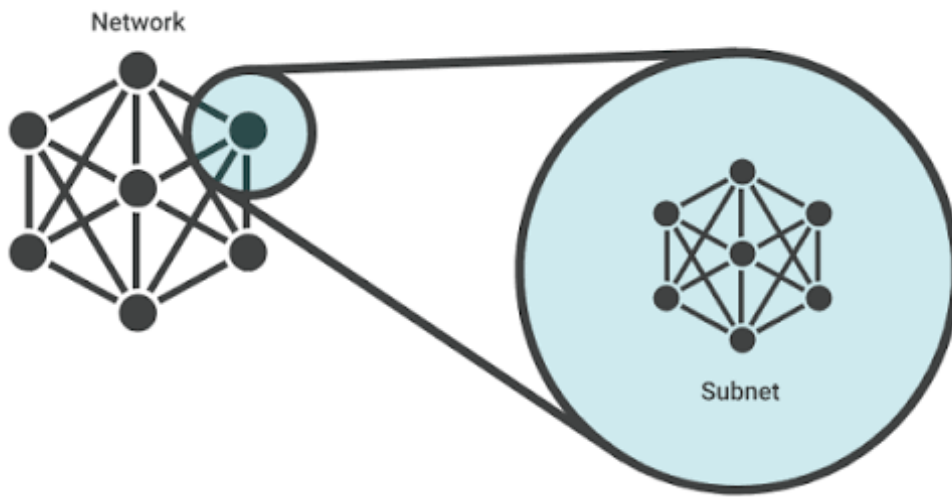


Class C 255.255.255.0 or /24
192.0.0.0 - 223.255.255.0



❖ What is subnet ?

- A subnet is a smaller network created by dividing a larger network into multiple parts.
- A subnet, or subnetwork, is a network inside a network. Subnets make networks more efficient. Through subnetting, network traffic can travel a shorter distance without passing through unnecessary routers to reach its destination.



❖ What is Subnet mask ?

A subnet mask is the number that tells devices which part of an IP address identifies the network and which part identifies the individual device (host).

A subnet mask becomes 32 bit number which identify –

- i) Who parts identify Network ID &
- ii) Who parts identify Host ID

In subnet mask

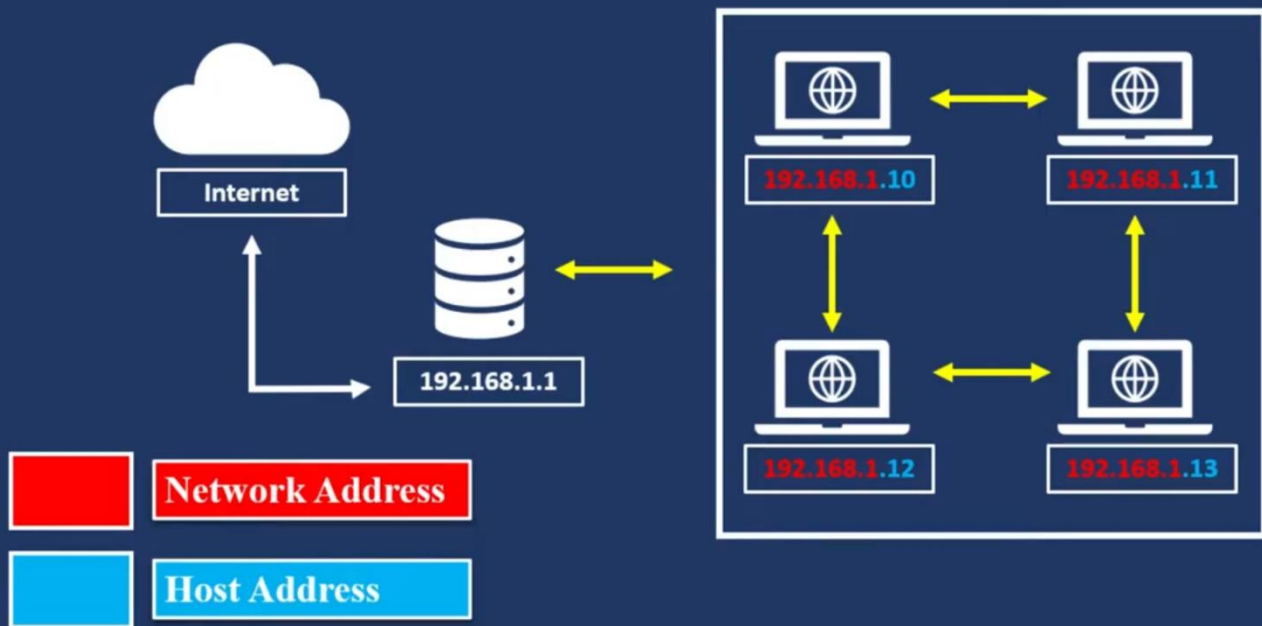
1 bit represent → network portion

0 bit represent → host portion

- Class C subnet mask 255.255.255.0 (convert in binary)

11111111	11111111	11111111	00000000
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Subnet Mask



❖ Default mask / Default Subnet mask of Classful IP

Address Class	Value in First Octet	Classful Mask (Dotted Decimal)	Classful Mask (Prefix Notation)
Class A	1–126	255.0.0.0	/8
Class B	128–191	255.255.0.0	/16
Class C	192–223	255.255.255.0	/24
Class D	224–239	—	—
Class E	240–255	—	—

IP address Classes

Class	# Network Bits	# Hosts Bits	Decimal Address Range	Subnet mask
Class A	8 bits	24 bits	1-126	255.0.0.0
Class B	16 bits	16 bits	128-191	255.255.0.0
Class C	24 bits	8 bits	192-223	255.255.255.0
Class D	Reserved for Multicasting		224-239	N/A
Class E	Reserved for R & D		240-255	N

❖ Classful IP Addressing Numerical Questions

If IP address : 192.168.10.70

Q1. Which Class / Identify the class ?

→ First octet = 192 (I know class C range = 192-223)

Therefore : Class = C

Q2. Find the default mask ?

→ Default mask of Class C is 255.255.255.0/24

Q3. Identify network and host

→ In Class C = 3 octet represent network and 1 octet represent host

So, 192.168.10.70

Network id = 192.168.10.0

Host id = 70

Note : In IP address, if ask network id then host part written 0 a/c to class.

Q4. What network address ?

→ Host part convert into 0.

Here ip address of Class C and Class C by default subnet mask /24 then

network address is 192.168.10.0

Q5. Find Broadcast address ?

→ if host portion all bits convert 1 then find broadcast address

192.168.10.255

This is broadcast address of ip address 192.168.10.70

Q6. Find useable host and host range ?

→ Except network id and broadcast addresss

Total useable host = $2^n - 2$ (where n is hosts bit)

= $2^8 - 2$ (here class c and this class hosts bit = 8)

= 256 - 2

= 254

Total host range = 192.168.10.1 – 192.168.10.254

Q6. Find 9th host IP id ?

→ first(1st) host ip = 192.168.10.1 then 9th host ip = 192.168.10.9

Q7. If ip 130.10.234.96 find 60,000th host ip address ?

→ $60,000 / 256 = 234$ (quotient) and 96 (remainder)

Here quotient write on 3rd octet and remainder write 4th octet

So, 60,000th host ip = 192.168.234.96

❖ Formula

✓ **Usable Hosts** = Total Host Addresses – 2

here 2 addresses are excluded because:

1. **Network Address** → Reserved to identify the network.

2. **Broadcast Address** → Reserved to communicate with all hosts in the network.

✓ **nth host ip address = $n/256$ (class B)**

Where, Third octet = quotient, Fourth octet = remainder

✓ **nth host ip address = $n/256$ (1st time) ,Again quotient/256 (2nd time) (class A)**

Where, first time divided → Fourth octet = remainder

Second time divided → second octet = quotient, third octet = remainder

Example : if ip 10.0.0.0 then find 6,00,000 usable host ip ?

→ $600000/256 = 2343$ (quotient), 192 (remainder)

Again $2343/256 = 9$ (quotient), 39 (remainder)

So, 600000th usable host ip = 10.9.39.192

❖ Total Network and Host/Device in Class A,B & C

Class	First Octet range	Default Subnet mask	Total Use Network	Total Hosts per network	Total usable Hosts per network = Total host - 2
A	1-126	255.0.0.0	126	$2^{24}=1,67,77,216$	1,67,77,214
B	128-191	255.255.0.0	16,382	$2^{16}=65,536$	65,534
C	192-223	255.255.255.0	20,97,150	$2^8=256$	254
D	224-239	N/A (Multicast)	-----	-----	-----
E	240-255	N/A (Exp. / research)	-----	-----	-----

Questions

Q1. Given IP: 10.0.0.0

Which class IP is this? Find the maximum number of hosts possible and maximum usable hosts.

Q2. How many networks are possible in **Class A**? Also find the number of usable hosts possible per network.

Q3. IP Address: 201.20.30.40

Find:

- Network ID
- 4th Host IP ID
- Last Host IP ID
- Broadcast Address

Q4. IP Address: 192.168.1.0

How many usable devices/hosts can be connected in this network using the default Class C subnet mask?

Q5. IP Address: 172.16.0.0

Which class is this? Find its total host capacity and usable host capacity.

Q6. How many networks are possible in **Class B**? Also find the usable hosts

possible per network.

Q7. How many networks are possible in **Class C**? Also find the usable hosts possible per network.

Q8. IP Address: 150.100.0.0

Which class does it belong to? Find the network size and host size.

Q9. IP Address: 200.10.10.0

Find:

- Network ID
- 1st Host IP
- Last Host IP
- Broadcast Address

Q10. If a **Class B** IP address is provided with its default subnet mask, how many usable hosts/devices are possible in one network?

Q11. IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Find the total number of usable hosts possible in this network.

Q12. IP Address: 25.10.20.30

Find:

- IP Class
- Default Subnet Mask
- Network ID
- 1st Host IP
- Last Host IP
- Broadcast Address

Q13. IP Address: 172.20.10.50

Find the **Class, Network ID, Host ID, First Host, Last Host and Broadcast Address** using classful addressing.

Q14. IP Address: 192.168.10.100

Find the **Network ID, First Usable Host, Last Usable Host and Broadcast Address** using the default Class C mask.

Q15. A company is given a **Class A** network 10.0.0.0.

Find:

- Number of network bits
- Number of host bits
- Total addresses
- Usable host addresses

Q16. A company is given a **Class B** network 172.16.0.0.

Find:

- Number of network bits
- Number of host bits
- Total addresses
- Usable host addresses

Q17. A company is given a **Class C** network 192.168.5.0.

Find:

- Number of network bits
- Number of host bits
- Total addresses
- Usable host addresses

Q18. IP Address: 100.20.30.40

Using **classful addressing**, find:

- Class
- Default Subnet Mask
- Network ID
- Host ID
- First Host IP
- Last Host IP
- Broadcast Address

Q19. IP Address: 180.50.10.25

Using the default classful subnet mask, find the **Network ID, First Host, Last Host and Broadcast Address.**

Q20. IP Address: 210.20.30.40

Find:

- IP Class
- Default Subnet Mask
- Network ID
- First Usable Host
- Last Usable Host
- Broadcast Address
- Maximum usable hosts in this network.

Q21. A company requires 680 usable ip addresses for its devices. Which class of IPv4 address should be assigned to the company under the classful addressing scheme ? (Ans = Class B)

Q22. A company requires 200 usable ip addresses for its devices. Which class of IPv4 address should be assigned to the company under the classful addressing scheme ? (Ans = class C)

Q23. A university needs IP addresses for 1,000 devices within a single network. If classful addressing is used and subnetting is not considered, which IP address class should be selected (Ans = Class B)

Answers

1. Class A; Total Hosts: 16,777,216; Usable Hosts: 16,777,214
2. 126 Networks; 16,777,214 Usable Hosts/Network
3. Network ID: 201.20.30.0
4th Host: 201.20.30.4
Last Host: 201.20.30.254
Broadcast: 201.20.30.255
4. 254 Usable Hosts
5. Class B; Total Hosts: 65,536; Usable Hosts: 65,534
6. 16,384 Networks; 65,534 Usable Hosts/Network
7. 2,097,152 Networks; 254 Usable Hosts/Network
8. Class B; Network Size: 65,536 Addresses
9. Network ID: 200.10.10.0
1st Host: 200.10.10.1
Last Host: 200.10.10.254
Broadcast: 200.10.10.255
10. 65,534 Usable Hosts/Devices
11. 254 Usable Hosts
12. Class A; Mask: 255.0.0.0; Network ID: 25.0.0.0; First Host: 25.0.0.1;
Last Host: 25.255.255.254; Broadcast: 25.255.255.255
13. Class B; Network ID: 172.20.0.0; Host ID: 10.50; First Host:
172.20.0.1; Last Host: 172.20.255.254; Broadcast: 172.20.255.255
14. Network ID: 192.168.10.0; First Host: 192.168.10.1; Last Host:
192.168.10.254; Broadcast: 192.168.10.255
15. Network Bits: 8; Host Bits: 24; Total Addresses: 16,777,216; Usable:
16,777,214
16. Network Bits: 16; Host Bits: 16; Total Addresses: 65,536; Usable:
65,534

17. Network Bits: 24; Host Bits: 8; Total Addresses: 256; Usable: 254
18. Class A; Mask: 255.0.0.0; Network ID: 100.0.0.0; Host ID: 20.30.40; First Host: 100.0.0.1; Last Host: 100.255.255.254; Broadcast: 100.255.255.255
19. Class B; Network ID: 180.50.0.0; First Host: 180.50.0.1; Last Host: 180.50.255.254; Broadcast: 180.50.255.255
20. Class C; Mask: 255.255.255.0; Network ID: 210.20.30.0; First Host: 210.20.30.1; Last Host: 210.20.30.254; Broadcast: 210.20.30.255; Usable Hosts: 254

❖ Problem/wastage ip address in IPv4 classful addressing

- From 1980-1993, In classful addressing, IPv4 networks were divided into fixed classes: A, B, and C.
- The problem was that each class had a fixed number of host addresses.
- For example:
 - Class A → 16,777,214 usable hosts
 - Class B → 65,534 usable hosts
 - Class C → 254 usable hosts
- Suppose an organization needs 500 IP addresses.
It cannot get a Class C because:
Class C → 254 usable IPs ❌
So it would need a Class B:
Class B → 65,534 usable IPs
Required → 500
Wasted → 65,034 IPs 😞

This caused IPv4 address wastage.

❖ Classless IP in IPv4.

- In IPv4, classful addressing caused IP address wastage. To solve this

problem, classless addressing (CIDR) was introduced in 1993.

- **Solution: Classless Addressing (CIDR)**
- **CIDR** = Classless Inter-Domain Routing

❖ What is CIDR ?

- CIDR is a classless IP addressing method that uses a prefix length (such as /24, /25, /26) to allocate IP addresses according to actual requirements.
- The classless concept was introduced, which is known as CIDR (Classless Inter-Domain Routing).
- In CIDR, the fixed Class A, Class B, and Class C concept is removed.
- Now, each IP network is represented using a prefix /n, where n represents the number of network bits. The remaining bits are used for the host portion.
- CIDR allows us to create smaller and more flexible subnets according to our requirements.

Example :

In Classful addressing: Class C $\rightarrow$ 255.255.255.0 $\rightarrow$ /24

Total 254 usable hosts

But suppose we need only **100 hosts**.

With CIDR, we can use:

192.168.1.0/25

Here:

/25 $\rightarrow$ 25 network bits

$\rightarrow$ 7 host bits

Total IPs = $2^7 = 128$

Usable = $128 - 2 = 126$

So we get **126 usable IP addresses**, instead of wasting a /24 network's 254 usable addresses.

❖ Rule of written classless Subnet Mask

In a subnet mask, 1s are always continuous, followed by continuous 0s.

Subnet mask (in Binary)	Wrong /Right	Reason
11111111.11111111.11111111.00000000	Right	Continuous 1 & 0
11111111.11111111.11111111.01111111	Wrong	No Continuous
11111111.11000000.00000000.00000000	Right	Continuous
11111111.11111111.00000000.11111111	Wrong	No Continuous
11110000.00000000.00000000.00000000	Right	Continuous
11111111.11111111.11111111.11100000	Right	Continuous
11111111.11100000.11100000.11111111	Wrong	No Continuous

Q.No. Binary Subnet Mask

Answer

- 1 11111111.11111111.11111111.10000000 ?
- 2 11111111.11111111.11111111.00111111 ?
- 3 11111111.11111111.11110000.00000000 ?
- 4 11111111.11111111.00001111.00000000 ?
- 5 11111111.11111111.11111110.00000000 ?
- 6 11111111.11111111.11111111.11110000 ?
- 7 11111111.11000000.00000000.00000000 ?
- 8 11111111.11111111.10111111.00000000 ?
- 9 11111111.11100000.00000000.00000000 ?
- 10 11111111.11111111.11111111.01000000 ?
- 11 11110000.00000000.00000000.00000000 ?
- 12 11111111.11111100.00000000.00000000 ?
- 13 11111111.11111111.11100000.00000000 ?
- 14 11111111.10000000.11100000.00000000 ?
- 15 11111111.11111111.11111111.11111110 ?

SUBNETTING

Subnet Chart				Decimal to Binary	
CIDR	Subnet Mask	Addresses	Wildcard	Subnet Mask	Wildcard
/32	255.255.255.255	1	0.0.0.0	255 1111 1111	0 0000 0000
/31	255.255.255.254	2	0.0.0.1	254 1111 1110	1 0000 0001
/30	255.255.255.252	4	0.0.0.3	252 1111 1100	3 0000 0011
/29	255.255.255.248	8	0.0.0.7	248 1111 1000	7 0000 0111
/28	255.255.255.240	16	0.0.0.15	240 1111 0000	15 0000 1111
/27	255.255.255.224	32	0.0.0.31	224 1110 0000	31 0001 1111
/26	255.255.255.192	64	0.0.0.63	192 1100 0000	63 0011 1111
/25	255.255.255.128	128	0.0.0.127	128 1000 0000	127 0111 1111
/24	255.255.255.0	256	0.0.0.255	0 0000 0000	255 1111 1111
/23	255.255.254.0	512	0.0.1.255		
/22	255.255.252.0	1,024	0.0.3.255		
/21	255.255.248.0	2,048	0.0.7.255		
/20	255.255.240.0	4,096	0.0.15.255		
/19	255.255.224.0	8,192	0.0.31.255		
/18	255.255.192.0	16,384	0.0.63.255		
/17	255.255.128.0	32,768	0.0.127.255		
/16	255.255.0.0	65,536	0.0.255.255		
/15	255.254.0.0	131,072	0.1.255.255		
/14	255.252.0.0	262,144	0.3.255.255		
/13	255.248.0.0	524,288	0.7.255.255		
/12	255.240.0.0	1,048,576	0.15.255.255		
/11	255.224.0.0	2,097,152	0.31.255.255		
/10	255.192.0.0	4,194,304	0.63.255.255		
/9	255.128.0.0	8,388,608	0.127.255.255		
/8	255.0.0.0	16,777,216	0.255.255.255		
/7	254.0.0.0	33,554,432	1.255.255.255		
/6	252.0.0.0	67,108,864	3.255.255.255		
/5	248.0.0.0	134,217,728	7.255.255.255		
/4	240.0.0.0	268,435,456	15.255.255.255		
/3	224.0.0.0	536,870,912	31.255.255.255		
/2	192.0.0.0	1,073,741,824	63.255.255.255		
/1	128.0.0.0	2,147,483,648	127.255.255.255		
/0	0.0.0.0	4,294,967,296	255.255.255.255		

Subnet Proportion	

Classful Ranges	
A	0.0.0.0 - 127.255.255.255
B	128.0.0.0 - 191.255.255.255
C	192.0.0.0 - 223.255.255.255
D	224.0.0.0 - 239.255.255.255
E	240.0.0.0 - 255.255.255.255

Reserved Ranges	
RFC1918	10.0.0.0 - 10.255.255.255
Localhost	127.0.0.0 - 127.255.255.255
RFC1918	172.16.0.0 - 172.31.255.255
RFC1918	192.168.0.0 - 192.168.255.255

Determine Usable Hosts	
Total Addresses	256
- Subnet ID	- 1
- Broadcast Address	- 1
Usable hosts	254

Terminology	
CIDR · Classless interdomain routing was developed to provide more granularity than legacy classful addressing; masks expressed in the form /XX are in CIDR notation	VLSM · Variable length subnet masks are an arbitrary length between 0 and 32 bits; CIDR relies on VLSMs to define routes

❖ Classless IP Addressing Numerical Questions

Q. Subnet mask 255.255.255.128 then how many usable hosts ?

- A. 126 B. 128 C. 124 D. 120

Method 1:

→ दिया है 255.255.255.128 (सब को यदि binary में बदले तो)

Binary of 255 = 11111111

Binary of 128 = 10000000

11111111.11111111.11111111.10000000 (सिर्फ 1 bit को count करें)

8.8.8.1 → इसे जोड़े दें → $8+8+8+1 = 25$ bit

Total bits 32 होता हैं तो $32 - 25 = 7$

$2^7 = 128$ host , usable host = $128 - 2 = 126$ ans

OR

Method 2:

Give → 255.255.255.128

255.255.255.(255-127)

$127 - 1 = 126$ ans (यहाँ 1 इसलिए कम किये क्योंकि network में 1 parity error पता करने के लिए करते हैं)

Q एक class C नेटवर्क 192.168.1.0 को /26(bit) subnet mask के साथ दिया गया हैं तो कितने usable hosts होंगे ?

A. 64 B. 62 C. 32 D. 30

→ कुल ipv4 में कुल bits 32, दिया हुआ हैं 26, तो $32-26 = 6$

अब, total host $2^6 = 64$, एवं usable host → $64-2 = 62$

Q. IP 10.0.0.0 and Subnet mask /8 then how many network create ?

A. 126 B. 254 C. 65,534 D. 1

→ ये class-A का ip हैं और class -A में network कुल 8 bits का होता हैं
तो total network → $8 - 8 = 0$, (यहाँ सिर्फ 1 network हैं)

Q3 How many total usable hosts in /30 subnet mask .

A. 4 B. 2 C. 6 D. 8

→ Give – 30 bits ($32 - 30 = 2$ bits)

Total host → $2^2 = 4$, Total Usable host = $4-2 = 2$ ans

Numerical Classless

Q1. IP 10.0.0.0/20 → total usable hosts

Q2. IP 172.16.0.0/18 → Total usable Ips

Q3. Network 192.168.1.0 /27 → first and last usable ip

Q4. Network 192.168.1.0/28 → How many subnet and total host in subnet.

Answer

Q1. $2^{12}-2 = 4094$

Q2. $2^{14}-2 = 16382$

Q3. First – 192.168.1.1 , Last – 192.168.1.30

Q4. Subnets $2^4 = 16$ and hosts $2^4-2 = 14$

❖ Private IP Address vs Public IP Address

Public IP Address

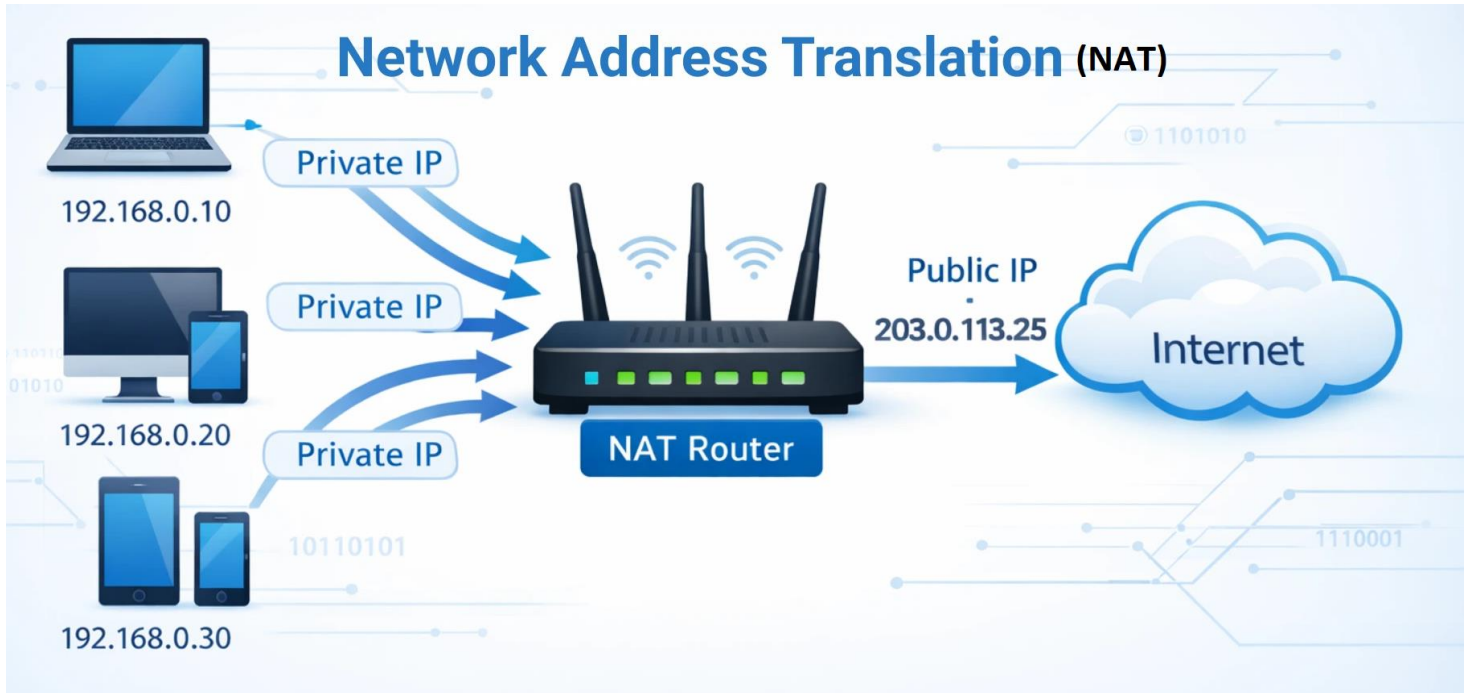
- A Public IP address is an IP address used for communication over the Internet.
- It must be globally unique at a given time.
- It is normally assigned to you by your Internet Service Provider (ISP).
- Your router may use the public IP to communicate with Internet servers.
- Example : 49.x.x.x, 103.x.x.x (here x = 0-255),

Private IP Address

- A Private IP address is used inside a local/private network (LAN), such as a home, school, or office network.
- It is not directly routable on the public Internet.
- Multiple devices can use private IP addresses in different networks.
- Usually assigned by a router using DHCP.
- IPv4 Private ip rang: (below fig)

Private IP address space	
From	To
10.0.0.0	10.255.255.255
172.16.0.0	172.31.255.255
192.168.0.0	192.168.255.255

❖ How to public ip to private ip or vice versa



=====HIRA KUMAR=====